A PostgreSQL Backend for Analysing Singapore HDB Resale Prices and Hawker-Centre Accessibility

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Abstract

This project develops a normalized PostgreSQL database linking HDB resale transactions to nearby hawker centres. The reproducible pipeline combines 2023–2025 resale records, current HDB coordinates, official hawker attributes, and supplied hawker coordinates. After validation and matching, the database-ready instance contains 26 towns, 9,385 HDB blocks, 78,670 transactions, 129 hawker centres, and 40,354 proximity links. We define three great-circle-distance bands and provide six analytical SQL queries plus candidate indexes. The complete schema, processed data, and all six analytical queries were executed successfully in PostgreSQL 18 through pgAdmin. The final all-table query returned 154 groups, matching the independent CSV cross-check.

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1 Introduction

Public housing and hawker centres are important parts of Singapore's social infrastructure. The project asks whether housing attributes, remaining lease, town, and accessibility to affordable food are associated with resale prices. We use official HDB transaction data (Housing and Development Board, 2026), official market and hawker attributes (Singapore Food Agency, 2026), and the geolocated HDB dataset described by Lan *et al.* (2022). The implementation follows relational normalization and integrity principles (Elmasri and Navathe, 2016).

2 Task 1: Conceptual and Logical Design

2.1 Proximity definition

We use haversine great-circle distance between the latitude/longitude of an HDB block and a hawker centre. The three mutually exclusive bands are: *near*, $[0, 500)$ m; *medium*, $[500, 1000)$ m; and *far*, $[1000, 2000]$ m. Links beyond 2 km are not stored because they do not represent useful local access. The exact distance is stored; its band is derived from a reference relation, avoiding two stored facts that can disagree. This choice is reproducible, requires no proprietary routing API, and is consistent with the coordinate datasets, although it approximates walking accessibility.

2.2 Conceptual ER model

Figure 1 follows the min–max convention taught in the SD6103 lectures (Nanyang Technological University, 2026a): rectangles are entity types, diamonds are relationship types, ellipses are attributes, underlining denotes keys, and a dashed ellipse denotes a derived attribute. The final model follows the assignment terminology **HDB_FLAT**, while defining its data-supported grain explicitly: one geocoded block–street address, not an individual dwelling unit. Thus `flat_id` identifies a block-address record. Introducing a second unit-level entity would invent identities absent from the sources. Flat type, storey range, floor area, model, lease and price instead describe a resale transaction observation.

A town may contain zero or many blocks, while every block belongs to exactly one town. A flat record may have zero or many resale transactions, while every transaction belongs to exactly one flat record through *has_sale*. The *nearby* relationship is M:N. Its `distance_m` is a relationship attribute and `proximity_class` is derived from that distance.

2.3 Synthesis of the candidate ERM designs

The seven proposed diagrams contain three recurring alternatives. Table 1 records the final decisions so that conceptual notation is not mixed with implementation details.

Table 1: Resolution of the candidate-model alternatives.

Issue	Selected design	Reason
Housing grain	One HDB_FLAT entity at block-address grain	No unit number exists; type, storey and area vary by transaction.
Block–hawker link	Direct M:N *nearby* relationship	This is the lecture's conceptual representation; `distance_m` is a relationship attribute.
Association relation	Composite key (block, hawker) in the logical model	Prevents duplicate pairs without an unnecessary surrogate `proximity_id`.
Proximity class	Derived, not stored in each pair	Prevents distance and class from contradicting one another.
Proximity levels	Logical reference relation	Centralizes the three interval definitions and supports controlled changes.
Fresh market flag	Derived Boolean attribute	PostgreSQL generates it from the official market-stall count, preventing contradictory values.

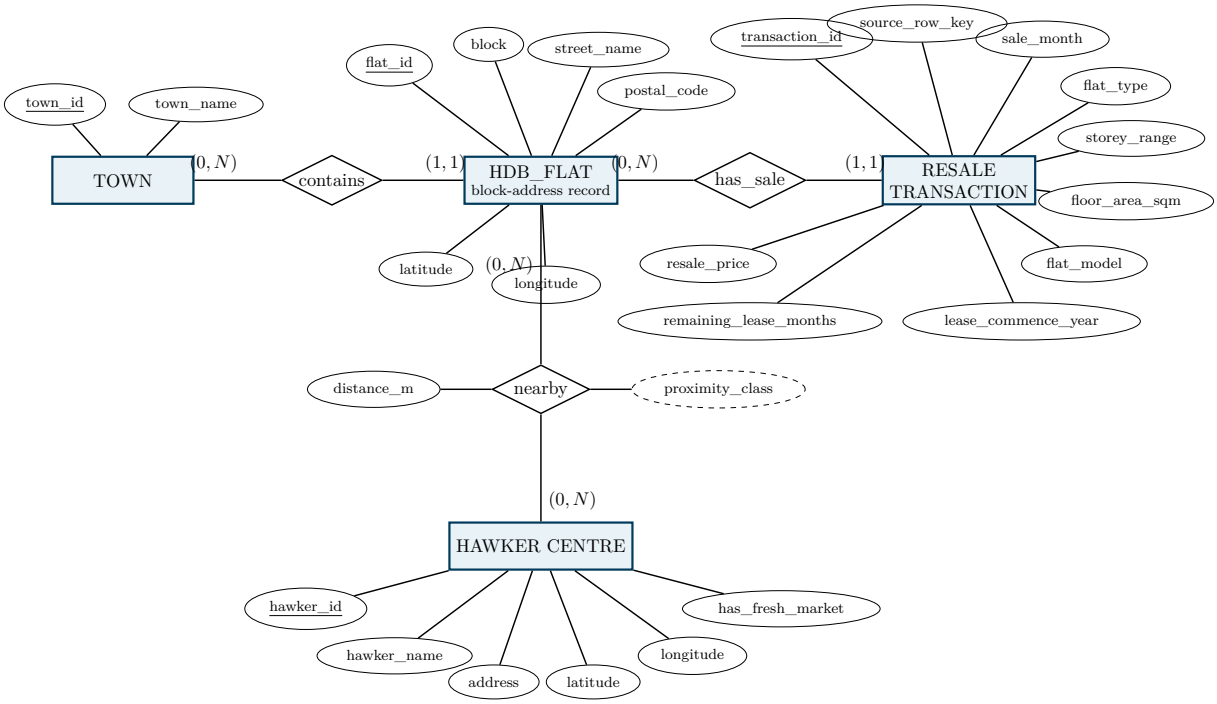


Figure 1: Final conceptual ER model. HDB_FLAT is a block-address record; proximity class is derived.

2.4 Relational schema and normalization

Relation	Attributes and constraints
Town	**town_id** (PK), town_name (UK, non-empty)
HDBFlat	**flat_id** (PK), town_id (FK, NOT NULL), block, street_name, postal_code, latitude, longitude; UK(block, street_name)
ResaleTransaction	**transaction_id** (PK), flat_id (FK, NOT NULL), sale_month, flat_type, storey_range, floor_area_sqm, flat_model, lease_commence_year, remaining_lease_months, resale_price, source_row_key (UK)
HawkerCentre	**hawker_id** (PK), hawker_name (UK), address, latitude, longitude, market_stall_count, has_fresh_market (generated)
ProximityLevel	**proximity_level_id** (PK), level_name (UK), min_distance_m, max_distance_m, upper_inclusive; interval overlap prohibited
Nearby	**flat_id, hawker_id** (composite PK/FKs), distance_m; relation produced by the M:N mapping

The 1:N *contains* and *has_sale* relationships are mapped by NOT NULL foreign keys on their N-sides. Following the lecture mapping rule for M:N relationships (Nanyang Technological University, 2026b), `nearby` contains both participant keys as a composite primary key and the relationship attribute `distance_m`. `ProximityLevel` is introduced at logical design time as a reference relation. The view `nearby_classified` derives the matching level rather than storing a redundant `proximity_level_id` in every pair.

For transactional correctness, the database makes the source-row hash unique, rejects non-positive prices and areas, constrains coordinates and lease values, cascades deletion only for derived flat–hawker links, and retains the default restrictive behavior for transaction history. PostgreSQL range exclusion prevents overlapping proximity-level definitions. During preparation, `has_fresh_market` is set from the official market-stall count; it remains `NULL` when no official match exists, preserving the distinction between "no market" and "not matched".

Every non-key attribute describes its relation's key, the whole key, and no other non-key attribute. Town names, proximity definitions, and hawker facts are separated, eliminating update anomalies; the schema is in 3NF and, under the declared candidate keys, BCNF. Domain checks constrain Singapore coordinates, positive areas/prices, monthly dates, postal-code format, lease bounds, and distance bounds. Complete executable DDL appears in Appendix A.

3 Task 2: Data Acquisition and Population

3.1 Sources and reproducibility

The download script retrieves the specified GitHub `hdb.csv` and the official HDB resale source, then retains transactions from 1 January 2023 through 31 December 2025, and the official hawker attribute table. The supplied `hawker_geo.csv` provides hawker coordinates. The transformation program writes one CSV per database table and computes proximity links before PostgreSQL bulk loading.

Table 3: Input and processed dataset sizes.

Dataset	Raw rows	Loaded/derived rows
HDB resale transactions	78,670 in scope	78,670
HDB buildings	12,442	9,385 participating blocks
Hawker attributes	107	104 postcode matches
Hawker coordinates	129	129 centres
Block–hawker pairs	–	40,354 within 2 km

3.2 Transformations

Text keys are trimmed and upper-cased; sale months are converted to first-of-month dates; numeric fields are parsed into typed values; and source rows receive SHA-256 keys for idempotent loading. The early resale file lacks a remaining-lease field, so remaining months are computed as

$$99 \times 12 - \big(12(y_s - y_l) + (m_s - 1)\big),$$

where (y_s, m_s) is the sale month and y_l is the lease commencement year. Values are bounded to $[0, 1188]$. Hawker attributes are joined to coordinates by six-digit postcode, which is more reliable than names containing abbreviations. The official market-produce stall count is retained, and PostgreSQL generates the fresh-market flag from it. Haversine distance is computed for every candidate block–hawker pair; links within 2 km are stored, while their bands are assigned only through the classification view.

3.3 Loading issues and resolutions

All 78,670 transactions in the specified 2023–2025 interval match a geocoded block; no in-scope transaction is excluded. Of 129 geolocated hawkers, 104 match official attributes by postcode; 81 have market-produce stalls, 23 do not, and 25 remain unknown. Seventeen geolocation rows have blank addresses, represented as SQL `NULL`. PostgreSQL staging tables and foreign-key joins enforce load order; `ON CONFLICT` makes repeat runs safe.

4 Task 3: Queries and Results

All six queries use one common housing-transaction scope: **1 January 2023 to 31 December 2025**, three complete calendar years and 36 months. The six logical base relations preserve the final ER entities and add one distance-band lookup: `town`, `hdb_flat`, `resale_transaction`, `hawker_centre`, `nearby`, and `proximity_level`. An HDB-flat record represents a geocoded block address, rather than an individually identified housing unit. The executable SQL includes explanatory trailing comments.

4.1 Derived proximity view and data scope

All queries use a lower bound of `2023-01-01` (inclusive) and an upper bound of `2026-01-01` (exclusive). The next-year boundary includes every month through December 2025. The same bounds appear in every query and in a table check constraint. Both stages of Q3 use the same period. All 36 calendar months are present; no matched transaction in this instance falls outside that interval.

The `nearby_classified` view derives near [0,500), medium [500,1000), and far [1000,2000] metres from stored great-circle distance. The final upper bound is inclusive. A numeric proximity rank preserves logical order. The view reads the three-row proximity lookup; the lookup is a logical implementation detail rather than a new conceptual ER entity. Its non-overlap constraint prevents one stored link matching two bands. Coordinates and hawker attributes remain the supplied reference snapshots, not historical reconstructions for each sale date.

```
CREATE OR REPLACE VIEW nearby_classified AS -- Classify stored distances from the shared lookup.
SELECT n.flat_id,n.hawker_id,n.distance_m, -- Preserve each relationship key and exact distance.
 p.proximity_level_id,p.level_name AS proximity_class, -- Expose the configured band.
 p.proximity_level_id AS proximity_rank -- Keep near, medium and far in numeric order.
FROM nearby n -- Read one row per address-centre pair.
JOIN proximity_level p ON n.distance_m>=p.min_distance_m -- Include the lower boundary.
 AND (n.distance_m<p.max_distance_m -- Exclude ordinary upper boundaries.
      OR (p.upper_inclusive AND n.distance_m<=p.max_distance_m)); -- Include the final 2 km
    endpoint.
```

Measure	Value	Definition
Raw sales in the selected period	78,670	January 2023 - December 2025
Matched transactions	78,670	No in-period geocode exclusions
hdb_flat records	9,385	Participating block-address records
Hawker centres	129	81 yes, 23 no, 25 unknown
nearby links	40,354	Retained pairs at or below 2,000 m
Observed sale months	36	January 2023 - December 2025

Table 4: Database instance used for all six queries; only 2023-2025 transactions are included.

All numerical results were executed in PostgreSQL 18.6 through PGlite 0.5.10 (WebAssembly), then independently recomputed with pandas. Figures 1–6 show actual result exports in an generated result-table viewer; they are not pgAdmin captures. Prices are nominal SGD. Native PostgreSQL timings and genuine pgAdmin captures are not established by this run. The project guide requires approval for alternative tools; such approval has not been established here.

4.2 Q1: Price per square metre across towns

The query selects 4 ROOM and 5 ROOM transactions from January 2023 through December 2025, then averages each sale's resale price divided by floor area. Each transaction has equal weight; the measure is not total price divided by total area. The result contains 52 town/type groups and 52,280 qualifying sales.

```
SELECT t.town_name, -- Identify the town.
       rt.flat_type, -- Separate 4 ROOM and 5 ROOM sales.
       round(avg(rt.resale_price/rt.floor_area_sqm),2) AS avg_price_per_sqm, -- Weight each sale
     equally.
       count(*) AS transaction_count -- Report the underlying sample size.
FROM resale_transaction rt -- Read the scoped transaction table.
JOIN hdb_flat f USING(flat_id) -- Attach the sale's address.
JOIN town t USING(town_id) -- Attach its town name.
WHERE rt.flat_type IN ('4 ROOM','5 ROOM') -- Keep the two required room types.
  AND rt.sale_month>=DATE '2023-01-01' -- Include January 2023 onward.
  AND rt.sale_month<DATE '2026-01-01' -- Include all of December 2025.
GROUP BY t.town_name,rt.flat_type -- Compute a mean for each town/type pair.
ORDER BY avg_price_per_sqm DESC,t.town_name,rt.flat_type; -- Rank groups deterministically.
```

Town	Type	Mean SGD/m2	Sales
Central Area	5 ROOM	11,680.14	60
Central Area	4 ROOM	10,714.41	211
Queenstown	4 ROOM	9,944.81	784
Queenstown	5 ROOM	9,254.12	186

Table 5: Q1: four highest of 52 groups, January 2023 - December 2025.

Q1 | Actual merged-query output | Jan 2023–Dec 2025
PostgreSQL 18.6 / PGlite 0.5.10 · 52 full-result rows

town name	flat type	avg price per sqm	transaction count
CENTRAL AREA	5 ROOM	11680.14	60
CENTRAL AREA	4 ROOM	10714.41	211
QUEENSTOWN	4 ROOM	9944.81	784
QUEENSTOWN	5 ROOM	9254.12	186
BUKIT MERAH	4 ROOM	9154.35	1244
TOA PAYOH	4 ROOM	9114.64	922
KALLANG/WHAMPOA	4 ROOM	8916.40	985
BUKIT TIMAH	5 ROOM	8828.14	45
BUKIT MERAH	5 ROOM	8482.83	518
TOA PAYOH	5 ROOM	8479.82	393

Figure 2: Q1 result excerpt: first ten of 52 groups in SQL output order; scope 2023-2025.

Insight. Central Area 5 ROOM has the highest mean at SGD 11,680.14/m2 (60 sales), followed by Central Area 4 ROOM at SGD 10,714.41/m2. This describes the selected three-year period. Lease, floor, flat model and building composition remain uncontrolled, so the ranking is not a causal measure of town value.

4.3 Q2: Remaining lease and transaction price

Each in-period sale enters one disjoint bucket: below 600 months, 600 to below 840 months, or at least 840 months. For each town/bucket, the query returns the continuous median price, mean price per square metre and volume. The 62 output groups partition all 78,670 transactions. Bucket order is numerical, rather than alphabetical.

```
WITH bucketed AS ( -- Assign each scoped sale to one remaining-lease bucket.
 SELECT f.town_id,rt.resale_price,rt.floor_area_sqm, -- Keep the values needed for aggregation.
  CASE WHEN rt.remaining_lease_months<600 THEN 1 -- Less than 50 years.
       WHEN rt.remaining_lease_months<840 THEN 2 -- At least 50 and less than 70 years.
       ELSE 3 END AS bucket_rank -- At least 70 years.
 FROM resale_transaction rt -- Read the shared three-year sample.
 JOIN hdb_flat f USING(flat_id) -- Attach the town identifier.
 WHERE rt.sale_month>=DATE '2023-01-01' -- Start the common period.
   AND rt.sale_month<DATE '2026-01-01' -- End after December 2025.
)
SELECT t.town_name, -- Name the town.
 CASE b.bucket_rank WHEN 1 THEN '< 50 years' -- Label the short-lease group.
                    WHEN 2 THEN '50 to < 70 years' -- Label the middle group.
                    ELSE '70+ years' END AS lease_bucket, -- Label the longest group.
 round(percentile_cont(0.5) WITHIN GROUP -- Use a continuous median.
```

```
  (ORDER BY b.resale_price)::numeric,2) AS median_resale_price, -- Report nominal SGD.
 round(avg(b.resale_price/b.floor_area_sqm),2) AS avg_price_per_sqm, -- Average per-sale ratios.
 count(*) AS transaction_volume -- Count each transaction once.
FROM bucketed b -- Aggregate the assigned buckets.
JOIN town t USING(town_id) -- Attach the town name.
GROUP BY t.town_name,b.bucket_rank -- Keep town and bucket populations separate.
ORDER BY t.town_name,b.bucket_rank; -- Use logical lease-bucket order.
```

Town	Lease bucket	Median SGD	Mean SGD/m2	Sales
Ang Mo Kio	< 50 years	436,944.00	5,995.75	14
Ang Mo Kio	50 to < 70 years	438,888.00	5,869.39	2,428
Ang Mo Kio	70+ years	900,000.00	9,280.82	634

Table 6: Q2: Ang Mo Kio excerpt from 62 groups, using only 2023-2025 sales.

Q2 | Actual merged-query output | Jan 2023–Dec 2025
PostgreSQL 18.6 / PGlite 0.5.10 · 62 full-result rows

town name	lease bucket	median resale price	avg price per sqm	transaction volume
ANG MO KIO	< 50 years	436944.00	5995.75	14
ANG MO KIO	50 to < 70 years	438888.00	5869.39	2428
ANG MO KIO	70+ years	900000.00	9280.82	634
BEDOK	< 50 years	385000.00	5828.79	125
BEDOK	50 to < 70 years	458000.00	5818.06	2901
BEDOK	70+ years	778000.00	7982.41	807
BISHAN	< 50 years	365000.00	5298.88	41
BISHAN	50 to < 70 years	788000.00	7645.85	978
BISHAN	70+ years	1010000.00	9469.67	119
BUKIT BATOK	50 to < 70 years	450000.00	5510.84	1758

Figure 3: Q2 result excerpt: first ten of 62 town/bucket groups, 2023-2025.

Insight. In Ang Mo Kio, the 70+ years group has a median of SGD 900,000 across 634 sales, compared with SGD 438,888 across 2,428 sales in the 50 to below 70 years group. The below-50 group has only 14 sales and a median of SGD 436,944, so its estimate is particularly sensitive to sample composition. Restricting the sale years reduces the earlier historical-price mixing, but differences in flat type, floor area and building cohort remain. The comparison does not isolate a lease effect.

4.4 Q3: Towns with large 3-room flats

The CTE ranks towns by the transaction-weighted mean area of 3 ROOM sales in 2023-2025 and selects ten, with town ID as a tie-break. The outer query keeps the same room type and date bounds, then finds each town's maximum price per calendar year. All three years are complete; the output contains 30 town/year groups.

```
WITH top_towns AS ( -- Rank towns using the same scoped 3 ROOM population.
 SELECT f.town_id,avg(rt.floor_area_sqm) AS avg_3room_floor_area_sqm -- Weight transaction
    observations.
 FROM resale_transaction rt -- Read in-period sales.
```

```
 JOIN hdb_flat f USING(flat_id) -- Attach the town identifier.
 WHERE rt.flat_type='3 ROOM' -- Rank only the required room type.
   AND rt.sale_month>=DATE '2023-01-01' -- Start the ranking period.
   AND rt.sale_month<DATE '2026-01-01' -- Include December 2025 in ranking.
 GROUP BY f.town_id -- Compute each town's mean area.
 ORDER BY avg_3room_floor_area_sqm DESC,f.town_id -- Break ties by a stable key.
 LIMIT 10 -- Keep exactly ten towns.
)
SELECT t.town_name, -- Identify each selected town.
 extract(year FROM rt.sale_month)::integer AS sale_year, -- Create calendar-year groups.
 round(x.avg_3room_floor_area_sqm,2) AS avg_3room_floor_area_sqm, -- Show why the town was
    selected.
 max(rt.resale_price) AS highest_resale_price -- Report the annual maximum for 3 ROOM.
FROM resale_transaction rt -- Read the scoped sale observations.
JOIN hdb_flat f USING(flat_id) -- Attach town membership.
JOIN top_towns x USING(town_id) -- Restrict to the ten selected towns.
JOIN town t USING(town_id) -- Attach town names.
WHERE rt.flat_type='3 ROOM' -- Keep the same type in the annual-maximum stage.
  AND rt.sale_month>=DATE '2023-01-01' -- Apply the same start in both stages.
  AND rt.sale_month<DATE '2026-01-01' -- Apply the same end in both stages.
GROUP BY t.town_name,sale_year,x.avg_3room_floor_area_sqm -- Keep each town/year separate.
ORDER BY x.avg_3room_floor_area_sqm DESC,t.town_name,sale_year; -- Present ranked towns and
    years.
```

Rank	Town	Mean area m2	2025 maximum SGD
1	Tampines	71.24	698,000.00
2	Ang Mo Kio	71.01	794,500.00
3	Bukit Timah	70.94	602,888.00
4	Jurong East	70.51	558,000.00
5	Bukit Panjang	70.37	548,888.00
6	Choa Chu Kang	69.81	530,000.00
7	Clementi	69.32	740,188.00
8	Yishun	68.84	620,000.00
9	Woodlands	68.71	588,000.00
10	Hougang	68.32	640,000.00

Table 7: Q3: ten selected towns and their complete-calendar-2025 maxima.

Q3 | Actual merged-query output | Jan 2023–Dec 2025
PostgreSQL 18.6 / PGlite 0.5.10 · 30 full-result rows

town name	sale year	avg 3room floor area sqm	highest resale price
TAMPINES	2023	71.24	620000.00
TAMPINES	2024	71.24	640000.00
TAMPINES	2025	71.24	698000.00
ANG MO KIO	2023	71.01	777888.00
ANG MO KIO	2024	71.01	780000.00
ANG MO KIO	2025	71.01	794500.00
BUKIT TIMAH	2023	70.94	570000.00
BUKIT TIMAH	2024	70.94	535000.00
BUKIT TIMAH	2025	70.94	602888.00
JURONG EAST	2023	70.51	495000.00

Figure 4: Q3 result excerpt from 30 rows: ten towns and three complete calendar years.

Insight. Tampines ranks first with a mean 3 ROOM area of 71.24 m2. The largest annual maximum among selected towns is Clementi in 2024, at SGD 840,000.00. Mean area and maximum price describe different features of the population. A yearly maximum is an exceptional sale, rather than a typical price.

4.5 Q4: Hawker-centre proximity effect

DISTINCT ON selects each address's nearest stored hawker, breaking distance ties by hawker ID. Nearest selection runs on the raw nearby relation before the band lookup, exposing its ordering to a potential composite index. The query compares near and far sales within the same town and flat type. It computes (mean near - mean far) x 100 / mean far before rounding, retaining groups with both populations. The 160 addresses with no retained centre within 2 km, covering 1,338 sales, are excluded rather than assigned to far. There are 73 comparable groups.

```
WITH nearest AS ( -- Select one physical centre before attaching the band lookup.
 SELECT DISTINCT ON (flat_id) flat_id,hawker_id,distance_m -- Keep one nearest link per address.
 FROM nearby -- Read the original relationship instead of multiplying it by sales.
 ORDER BY flat_id,distance_m,hawker_id -- Match the useful B-tree order and break distance ties.
), prices AS ( -- Compare the same town and room type in the common date window.
 SELECT t.town_name,rt.flat_type, -- Define each comparison group.
  avg(rt.resale_price) FILTER (WHERE p.level_name='near') AS avg_near, -- Mean within 500 metres.

  avg(rt.resale_price) FILTER (WHERE p.level_name='far') AS avg_far -- Mean at 1,000-2,000
    metres.
 FROM resale_transaction rt -- Read sale observations at transaction grain.
 JOIN hdb_flat f USING (flat_id) -- Attach the sale's block address.
 JOIN town t USING (town_id) -- Attach its town name.
 JOIN nearest n USING (flat_id) -- Count each covered sale only once.
 JOIN proximity_level p ON n.distance_m>=p.min_distance_m -- Include the lower band boundary.
  AND (n.distance_m<p.max_distance_m -- Exclude ordinary upper boundaries.
       OR (p.upper_inclusive AND n.distance_m<=p.max_distance_m)) -- Include the final 2 km
    endpoint.
 WHERE rt.sale_month>=DATE '2023-01-01' -- Start the agreed three-year analysis scope.
   AND rt.sale_month<DATE '2026-01-01' -- Include all of December 2025.
 GROUP BY t.town_name,rt.flat_type -- Preserve separate comparison populations.
)
SELECT town_name AS town,flat_type, -- Match the required five-column output schema.
 round(avg_near,2) AS avg_price_near_hawker, -- Display near prices in nominal SGD.
 round(avg_far,2) AS avg_price_far_hawker, -- Display far prices in nominal SGD.
 round((avg_near-avg_far)*100/NULLIF(avg_far,0),2) AS price_diff_percentage -- Use unrounded
    means.
FROM prices -- Read the already grouped comparison values.
WHERE avg_near IS NOT NULL AND avg_far IS NOT NULL -- Require both near and far observations.
ORDER BY price_diff_percentage DESC,town,flat_type; -- Present deterministic ranked differences.
```

Town	Type	Near SGD	Far SGD	Diff. pct
Kallang/Whampoa	4 ROOM	860,820.60	575,672.00	49.53
Geylang	4 ROOM	789,293.64	565,056.25	39.68
Geylang	5 ROOM	867,796.36	640,000.00	35.59
Toa Payoh	5 ROOM	1,010,254.57	749,398.00	34.81
Toa Payoh	4 ROOM	853,958.44	645,219.03	32.35

Table 8: Q4: five largest differences among 73 comparable groups, 2023-2025.

Q4 | Actual merged-query output | Jan 2023–Dec 2025
PostgreSQL 18.6 / PGlite 0.5.10 · 73 full-result rows

town	flat type	avg price near hawker	avg price far hawker	price diff percentage
KALLANG/WHAMPOA	4 ROOM	860820.60	575672.00	49.53
GEYLANG	4 ROOM	789293.64	565056.25	39.68
GEYLANG	5 ROOM	867796.36	640000.00	35.59
TOA PAYOH	5 ROOM	1010254.57	749398.00	34.81
TOA PAYOH	4 ROOM	853958.44	645219.03	32.35
PUNGGOL	5 ROOM	842557.69	670286.55	25.70
BUKIT PANJANG	EXECUTIVE	838081.76	669789.24	25.13
TAMPINES	2 ROOM	435000.00	355416.67	22.39
TAMPINES	5 ROOM	867023.21	721820.21	20.12
KALLANG/WHAMPOA	5 ROOM	964237.92	809680.50	19.09

Figure 5: Q4 result excerpt: first ten of 73 groups by percentage difference.

Insight. Kallang/Whampoa 4 ROOM has the largest positive difference: near SGD 860,820.60, far SGD 575,672.00, or 49.53%. Across the 73 groups, 41 differences are positive and 32 negative. The mixed signs do not support a universal near-hawker premium. Housing mix, lease and other neighbourhood amenities remain potential confounders.

4.6 Q5: Prices near hawkers with fresh-market stalls

An address qualifies for a class when it has at least one hawker with a positive fresh-market stall count in that class. Unknown status is excluded. The CTE makes each address/class pair unique before joining sales, so several qualifying centres do not multiply a sale within a group. The budget is fixed at **SGD 700,000** before timing and is unchanged across all five sizes. This three-year sample gives 11 passing town/class groups from 5 of 22 eligible towns. SGD 350,000 would select all 22 eligible towns in this sample.

```
WITH eligible AS ( -- Avoid giving extra weight to multiple fresh-market centres.
 SELECT DISTINCT nc.flat_id,nc.proximity_class,nc.proximity_rank -- One address/class membership.

 FROM nearby_classified nc -- Read classified address-centre links.
 JOIN hawker_centre h USING(hawker_id) -- Attach market-stall status.
 WHERE h.has_fresh_market IS TRUE -- Exclude confirmed no and unknown status.
)
SELECT t.town_name,e.proximity_class, -- Group by town and distance band.
 round(avg(rt.resale_price),2) AS avg_resale_price, -- Average each within-group sale once.
 count(*) AS transaction_count -- Report the within-group sample size.
FROM resale_transaction rt -- Read the three-year sale observations.
JOIN hdb_flat f USING(flat_id) -- Attach town membership.
JOIN town t USING(town_id) -- Attach town names.
JOIN eligible e USING(flat_id) -- Use deduplicated fresh-market memberships.
WHERE rt.sale_month>=DATE '2023-01-01' -- Start the common analysis window.
  AND rt.sale_month<DATE '2026-01-01' -- Include the end of December 2025.
GROUP BY t.town_name,e.proximity_class,e.proximity_rank -- Keep distance groups separate.
HAVING avg(rt.resale_price)>700000 -- Use one fixed SGD 700,000 budget at every size.
ORDER BY avg_resale_price DESC,t.town_name,e.proximity_rank; -- Rank passing groups.
```

Town	Class	Mean SGD	Sales
Bukit Timah	near	897,986.20	162
Bishan	medium	848,219.63	886
Bishan	far	823,020.47	1,138
Central Area	near	805,810.01	450
Central Area	medium	787,059.31	495
Central Area	far	760,729.39	546
Bukit Merah	near	731,309.10	2,656
Bukit Merah	far	728,198.21	2,957
Bukit Merah	medium	717,838.37	2,424
Queenstown	far	712,094.01	1,902
Queenstown	medium	704,915.90	1,805

Table 9: Q5: all 11 groups above the fixed SGD 700,000 budget, 2023-2025.

Size	Eligible towns	Passing towns	Groups
S1	22	4	9
S2	22	5	10
S3	22	5	11
S4	22	5	10
S5	22	5	11

Table 10: Q5: every measured size is nonempty and selects fewer than all eligible towns.

Q5 | Actual merged-query output | Jan 2023–Dec 2025
PostgreSQL 18.6 / PGlite 0.5.10 · 11 full-result rows

town name	proximity class	avg resale price	transaction count
BUKIT TIMAH	near	897986.20	162
BISHAN	medium	848219.63	886
BISHAN	far	823020.47	1138
CENTRAL AREA	near	805810.01	450
CENTRAL AREA	medium	787059.31	495
CENTRAL AREA	far	760729.39	546
BUKIT MERAH	near	731309.10	2656
BUKIT MERAH	far	728198.21	2957
BUKIT MERAH	medium	717838.37	2424
QUEENSTOWN	far	712094.01	1902

Figure 6: Q5 result excerpt: first ten of 11 passing groups, 2023-2025.

Insight. Bukit Timah near has the highest passing mean, SGD 897,986.20, while Bishan medium and far also exceed SGD 800,000. High-price groups occur in different bands. A sale can belong to more than one class; group volumes therefore cannot be added to obtain a unique-sale total.

4.7 Q6: All-relation join

The additional question asks how address coverage, transaction volume and mean price differ by town, distance band and fresh-market status. All six base relations participate explicitly, including the proximity lookup. Nearby, hawker and proximity memberships are deduplicated before joining town, address and transaction rows. Unique address/class/status memberships are formed before transactions are joined. The result contains 154 groups, including 57 groups with unknown market status. An address or sale may occur in several different groups, but only once within each group.

```
WITH membership AS ( -- Deduplicate small spatial memberships before joining transactions.
 SELECT DISTINCT n.flat_id,p.proximity_level_id AS proximity_rank, -- Keep one address/band key.
  p.level_name AS proximity_class, -- Use the teammate's shared band definitions.
  CASE h.has_fresh_market WHEN TRUE THEN 'yes' -- Preserve confirmed fresh-market access.
                          WHEN FALSE THEN 'no' -- Preserve confirmed absence.
                          ELSE 'unknown' END AS fresh_market_status -- Keep missing evidence
    separate.
 FROM nearby n -- Include the M:N relationship base relation.
 JOIN hawker_centre h USING (hawker_id) -- Include the centre base relation.
 JOIN proximity_level p ON n.distance_m>=p.min_distance_m -- Include the lookup base relation.
  AND (n.distance_m<p.max_distance_m -- Apply non-overlapping interval boundaries.
       OR (p.upper_inclusive AND n.distance_m<=p.max_distance_m)) -- Retain the inclusive last
    endpoint.
)
SELECT t.town_name,m.proximity_class,m.fresh_market_status, -- Define the reporting populations.
 count(DISTINCT f.flat_id) AS flat_count, -- Count participating block-address records.
 count(*) AS transaction_count, -- Count each sale once per unique membership group.
 round(avg(rt.resale_price),2) AS avg_resale_price -- Use the same population as the count.
FROM town t -- Include the town base relation.
JOIN hdb_flat f USING (town_id) -- Include the address base relation.
JOIN resale_transaction rt USING (flat_id) -- Include the sale base relation: six in total.
JOIN membership m USING (flat_id) -- Use deduplicated memberships, avoiding hawker fan-out.
WHERE rt.sale_month>=DATE '2023-01-01' -- Start the same period used by Q1-Q5.
  AND rt.sale_month<DATE '2026-01-01' -- Include all three complete calendar years.
GROUP BY t.town_name,m.proximity_class,m.proximity_rank,m.fresh_market_status -- Separate
    statuses.
ORDER BY t.town_name,m.proximity_rank,m.fresh_market_status; -- Preserve logical distance order.
```

Town	Class	Status	Addresses	Sales	Mean SGD
Bukit Batok	near	unknown	49	1,598	605,640.12
Bukit Batok	medium	unknown	112	801	563,789.31
Bukit Batok	far	unknown	225	1,988	572,255.17

Table 11: Q6: Bukit Batok / unknown, all three distance groups within 2023-2025.

Q6 | Actual merged-query output | Jan 2023–Dec 2025
PostgreSQL 18.6 / PGlite 0.5.10 · 154 full-result rows

town name	proximity class	fresh market status	flat count	transaction count	avg resale price
BUKIT BATOK	near	unknown	49	1598	605640.12
BUKIT BATOK	medium	unknown	112	801	563789.31
BUKIT BATOK	far	unknown	225	1988	572255.17

Figure 7: Q6 filtered excerpt: the same Bukit Batok / unknown groups shown in the table.

Insight. Bukit Batok near/unknown contains 49 participating addresses and 1,598 sales, with mean SGD 605,640.12. Treating unknown as confirmed no would mislabel this population. Q6 pools all room types and accepts any matching centre in the band; Q4 uses a single nearest centre and separates room types. Their means therefore describe different populations, even with the same three-year date filter.

4.8 Merged-code performance measurements

All five increasing sizes use the same 2023–2025 period. A stable source-row hash assigns each sale to one of five disjoint partitions: the first sixteen hexadecimal digits, interpreted as an integer, modulo five, plus one. The next partition is added at each size. All 36 months occur at every size; S5 contains every scoped transaction exactly once. Five reference relations remain fixed, including the three-row proximity lookup.

Protocol scope. The project guide's original five historical source-file increments differ from these within-period partitions. All selected 2023–2025 sales originate in the latest historical file. Approval of this revised protocol, or an additional experiment with the original five-file procedure, is required before claiming full compliance with that requirement.

Size	Transactions	Months	Six-table MiB
S1	15,640	36	9.68
S2	31,581	36	14.88
S3	47,295	36	19.48
S4	63,007	36	24.39
S5	78,670	36	29.03

Table 12: Measured database footprints, including constraint indexes and TOAST, before analytical indexes.

Each reference relation is bulk-loaded once. VACUUM (ANALYZE) precedes every transaction-size baseline; data and maintenance state remain fixed throughout Task 4. Primary-key, unique and exclusion indexes are retained. No optional analytical index exists in Task 3. Each complete SELECT is exported, warmed up once, and executed five measured times. Query order rotates cyclically. The metric is PostgreSQL Execution Time from EXPLAIN (ANALYZE, BUFFERS, TIMING OFF, FORMAT JSON); planning is recorded separately. Loading, output transfer and rendering are outside timing. No COUNT wrapper replaces the required SELECT.

Size	Q1	Q2	Q3	Q4	Q5	Q6
S1	14.81	29.21	12.69	51.14	61.52	196.72
S2	25.16	56.42	20.66	59.77	86.34	348.50
S3	38.88	85.84	25.85	72.39	87.25	495.69
S4	53.01	128.96	34.03	87.39	100.27	705.80
S5	61.13	160.22	43.31	92.10	116.93	805.42

Table 13: Mean executor milliseconds for the 150 fresh merged-code Task 3 observations.

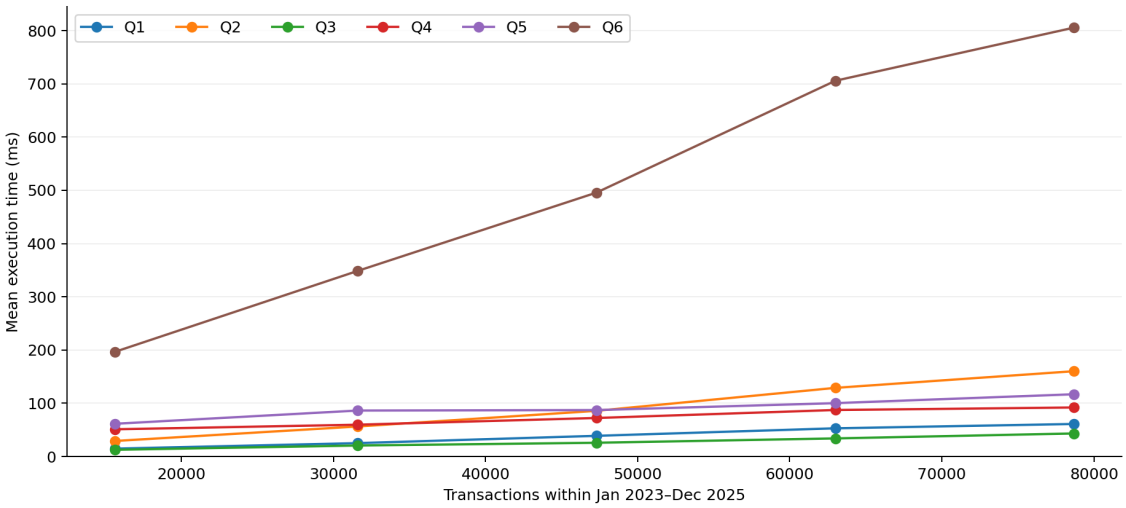


Figure 8: Fresh baseline means across five fixed-period cumulative populations.

Insight. Q6 is the largest cost in this execution environment because its valid address/band/status memberships still expand into many sale memberships before grouping. Deduplicating membership before the transaction join protects correctness and reduces intermediate multiplication, but does not make the remaining aggregate selective. Sequential scans are reasonable for queries consuming most of the scoped population. These are measured scaling observations, not native disk-I/O or cross-machine claims.

4.9 Integration and correctness checks

The integration uses the teammate's six-relation lookup design and bulk-load approach, together with the scoped package's row keys, date constraints, nullable fresh-market status, fixed Q5 budget and independent validation. The uploaded SQL contributes the transaction-deduplication principle; the final Q5/Q6 form applies equivalent deduplication at the smaller address-membership grain. Q4 keeps deterministic nearest-centre selection. A single query-file set supplies displayed SQL, exported results and every benchmark configuration.

Independent pandas recomputation passes all six largest-data outputs: 52, 62, 30, 73, 11 and 154 groups. Every optional-index configuration produces the exact same complete output fingerprint as the S5 baseline. The common scope is 78,670 transactions and 36 months, with 40,354 stored and classified nearby links. The prior historical native timings and the uploaded 1997–1999 Q1 window are not reused as results for this instance.

5 Task 4: Optimised Index Selection and Runtime Evaluation

5.1 Selection basis and controlled protocol

Five Task 4 responses were reconciled before choosing the final index bundle. Results from the all-history database and the PGlite/WASM experiment are useful sensitivity checks, but the primary evidence is the native PostgreSQL 18.4 experiment that uses the same 78,670 transactions, January 2023–December 2025 scope, schema and Q1–Q6 definitions as Tasks 1–3. Each condition received one excluded warm-up and five measured executions; all complete outputs were equal before and after index creation. The metric is server execution time from `EXPLAIN (ANALYZE, TIMING OFF)`. No scan type was forced.

The selection rule retains an index only when it supplies a required filter, join or order and its benefit is visible in an actual plan or in a statistically clear bundle effect. It rejects duplicate indexes and indexes that merely classify every row. On this basis the final solution contains four indexes and deliberately excludes an index on `remaining_lease_months`.

5.2 Final four-index solution

```
SET search_path = housing, public; -- Apply the measured four-index bundle.

CREATE INDEX idx_rt_type_month
ON resale_transaction (flat_type, sale_month)
INCLUDE (flat_id, floor_area_sqm, resale_price); -- Cover Q1 and both Q3 scans.

CREATE INDEX idx_rt_flat
ON resale_transaction (flat_id)
INCLUDE (transaction_id, resale_price); -- Support flat-led transaction lookups in Q4-Q6.

CREATE INDEX idx_nearby_flat_dist
ON nearby (flat_id, distance_m, hawker_id); -- Supply Q4's nearest-centre order.

CREATE INDEX idx_nearby_hawker
ON nearby (hawker_id)
INCLUDE (flat_id, distance_m); -- Cooperate with idx_rt_flat to remove Q6's spilling sort.

VACUUM (ANALYZE) resale_transaction; -- Equalise visibility and refresh planner statistics.
VACUUM (ANALYZE) nearby; -- Refresh statistics for both proximity access paths.

-- Deliberately omitted: an index on remaining_lease_months. Q2 classifies every
-- transaction with CASE and therefore cannot use that index to eliminate rows.
```

Index	Target	Plan-level purpose
`idx_rt_type_month`	Q1, Q3	Type-led filtering plus a month range; included columns permit narrow index-only access for price/area calculations.
`idx_rt_flat`	Q4–Q6	Fetch transactions by the flat foreign key; cooperates with the reverse nearby index in Q6.
`idx_nearby_flat_dist`	Q4	Provides `(flat_id,distance_m,hawker_id)` order for deterministic nearest-centre selection.
`idx_nearby_hawker`	Q6	Provides the reverse M:N access path; together with `idx_rt_flat`, removes Q6's spilling sort.

Table 14: Final indexes and the operators they are intended to change.

The four indexes occupy 9.27 MiB in the measured instance. This is a deliberate read-performance trade-off: future inserts must maintain four additional structures, so indexes without demonstrated benefit are not retained.

5.3 Measured results on the matching 2023–2025 database

Query	Base mean	Base SD	Index mean	Index SD	Reduction %
Q1	28.83	0.49	22.13	0.07	23.24
Q2	75.11	5.02	76.92	16.70	-2.41
Q3	21.42	6.43	9.08	0.93	57.64
Q4	36.74	0.86	30.50	5.19	16.98
Q5	236.08	38.88	220.29	10.17	6.69
Q6	992.47	22.35	480.42	7.14	51.59

Table 15: Native PostgreSQL mean execution time and sample SD in milliseconds; five repetitions per condition.

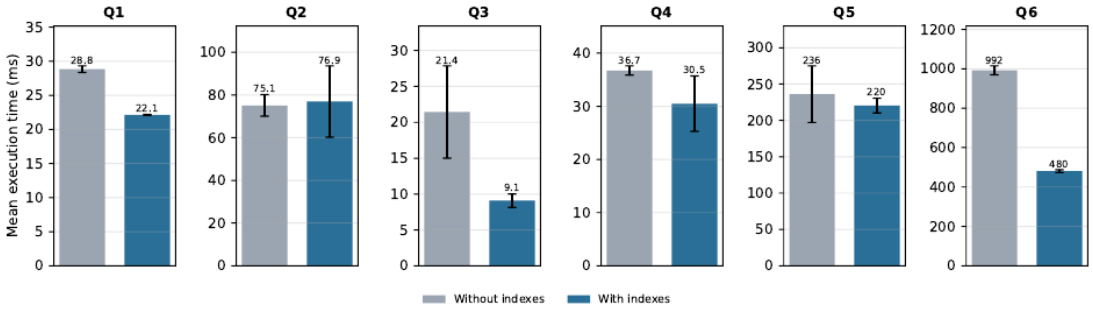


Figure 9: Native PostgreSQL baseline versus the optimised four-index bundle. The source chart reports the mean and one sample standard deviation for five runs per condition.

The sum of the six means falls from 1,390.65 ms to 839.34 ms, a 39.64% workload reduction. This aggregate is driven by large, repeatable gains in Q3 and Q6 rather than by treating every small timing difference as meaningful.

5.4 Per-query interpretation

Q1. The type/month covering index reduces execution time by 23.24%. Since the stored database itself contains only 2023–2025, the date predicate covers the full table; the measured gain comes from the flat-type predicate and the narrower covering structure. A longer history would make the month range additionally selective.

Q2. The 2.41% slowdown is within run-to-run noise. Q2 applies a CASE classification and median calculation to every transaction, so a single-column lease index cannot eliminate rows. Consequently `remaining_lease_months` is not indexed.

Q3. Both 3 ROOM scans switch from a full-table path to the covering type/month index. The mean falls by 57.64%, the largest proportional improvement.

Q4. Ordering nearby rows by flat, distance and hawker ID removes the sort before `DISTINCT ON`. The 16.98% reduction is useful but smaller than Q1/Q3 because the transaction join remains.

Q5. Its 6.69% reduction is not larger than the observed timing noise. The dominant work is membership de-duplication and grouping, so no additional Q5-specific index is claimed.

Q6. The reverse nearby path and flat-led transaction lookup must be used together. They change the join order and eliminate the approximately 16.3 MiB external sort observed in the baseline. Mean runtime falls from 992.47 to 480.42 ms, a 51.59% reduction. Separate trials found

that the flat index alone saved only about 14%, while the reverse nearby index alone was unused; the interaction is therefore part of the final design.

5.5 Alternatives rejected

The former three-index bundle is not retained. Its lease index is unused by Q2, and its simple month/type definition omits covering columns and uses the weaker leading-column order for this scoped database. Partial 3 ROOM and 4/5 ROOM indexes remain reasonable future candidates, but their current-dataset isolated tests did not outperform the confirmed native four-index bundle. Likewise, full-history runtimes are not mixed with the 2023–2025 measurements because their selectivity, row counts and dominant operators differ.

The delivered `sql/04_indexes.sql` is therefore the final operational answer: retain the four indexes shown above; do not create a lease index; and run equal `VACUUM (ANALYZE)` maintenance before any new baseline/index comparison.

5.6 References and provenance

PostgreSQL 18 documentation: https://www.postgresql.org/docs/18/sql-explain.html; index-only scans, https://www.postgresql.org/docs/18/indexes-index-only-scans.html; partial indexes, https://www.postgresql.org/docs/18/indexes-partial.html.

Teammate source: https://github.com/yzxfwdplf/SD6103-Data-Systems-Task3. The archived snapshot records file-level blob SHAs and URLs, with the root tree labelled separately. Uploaded SQL, scoped-data manifests and validation evidence are preserved in the package.

6 Limitations and Conclusion

Straight-line distance ignores street networks, crossings, and public transport. Current HDB coordinates are a current reference snapshot rather than historical coordinates, and hawker matching leaves 25 unknown market statuses. Prices are nominal rather than inflation-adjusted. Despite these limitations, the deliverable provides a normalized, constraint-rich, operational PostgreSQL backend and readable SQL for all required questions. Schema creation, five-file bulk loading, integrity checks, indexing, six analytical queries, and the controlled performance experiment have all been verified in PostgreSQL 18.

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A Database Creation SQL

```
BEGIN; -- Create one consistent six-relation instance.
CREATE SCHEMA housing; -- Use a new database or a fresh dedicated schema.
SET search_path = housing, public; -- Resolve unqualified project table names.
CREATE TABLE town ( -- Store town names once.
 town_id integer PRIMARY KEY, -- Preserve stable source town identifiers.
 town_name varchar(80) NOT NULL UNIQUE -- Prevent duplicate town names.
);
CREATE TABLE hdb_flat ( -- Represent one geocoded block-address record.
 flat_id bigint PRIMARY KEY, -- Map the conceptual flat key to the audited block key.
 town_id integer NOT NULL REFERENCES town(town_id), -- Assign each address to one town.
 block varchar(16) NOT NULL, -- Retain the block number, including letter suffixes.
 street_name varchar(120) NOT NULL, -- Retain the normalized street name.
 postal_code char(6), -- Preserve nullable postal codes as text.
 latitude numeric(9,6) NOT NULL, -- Use the audited HDB coordinate precision.
 longitude numeric(9,6) NOT NULL, -- Use the audited HDB coordinate precision.
 UNIQUE(block,street_name), -- Permit only one row per block and street.
 CHECK(postal_code IS NULL OR postal_code ~ '^[0-9]{6}$') -- Reject invalid nonempty postcodes.
);
CREATE TABLE resale_transaction ( -- Store a sale observation separately from its address.
 transaction_id bigint PRIMARY KEY, -- Keep the original transaction identifier.
 flat_id bigint NOT NULL REFERENCES hdb_flat(flat_id), -- Link each sale to one address.
 sale_month date NOT NULL, -- Store the first day of each sale month.
 flat_type varchar(20) NOT NULL, -- Keep room type at transaction grain.
 storey_range varchar(20) NOT NULL, -- Keep the reported storey interval.
 floor_area_sqm numeric(7,2) NOT NULL CHECK(floor_area_sqm>0), -- Require positive area.
 flat_model varchar(60) NOT NULL, -- Preserve the source model description.
 lease_commence_year smallint NOT NULL, -- Keep the declared lease start year.
 remaining_lease_months integer NOT NULL CHECK(remaining_lease_months BETWEEN 0 AND 1188), --
    Use a common month unit.
 resale_price numeric(12,2) NOT NULL CHECK(resale_price>0), -- Require a positive nominal SGD
    amount.
 source_row_key char(64) NOT NULL UNIQUE, -- Preserve distinct source observations.
 CHECK(sale_month >= DATE '2023-01-01' AND sale_month < DATE '2026-01-01'), -- Enforce the
    common 36-month scope.
 CHECK(sale_month=date_trunc('month',sale_month)::date) -- Require a first-of-month date.
);
CREATE TABLE hawker_centre ( -- Retain the supplied reference snapshot.
 hawker_id integer PRIMARY KEY, -- Preserve the source hawker key.
 hawker_name varchar(200) NOT NULL UNIQUE, -- Store one centre name per key.
 address varchar(300), -- Keep missing addresses as NULL.
 latitude numeric(12,9) NOT NULL, -- Preserve the source coordinate precision.
 longitude numeric(12,9) NOT NULL, -- Preserve the source coordinate precision.
 market_stall_count smallint CHECK(market_stall_count>=0), -- Keep unknown counts nullable.
 has_fresh_market boolean GENERATED ALWAYS AS ( -- Derive status from the source stall count.
  CASE WHEN market_stall_count IS NULL THEN NULL -- Preserve unknown status.
       ELSE market_stall_count>0 END -- Distinguish confirmed yes from confirmed no.
 ) STORED -- Prevent independent edits that contradict the count.
);
CREATE TABLE proximity_level ( -- Keep the teammate's configurable three-band lookup.
 proximity_level_id smallint PRIMARY KEY, -- Use the band key as its display rank.
 level_name varchar(20) NOT NULL UNIQUE, -- Name each distance class once.
 min_distance_m integer NOT NULL, -- Store the inclusive lower boundary in metres.
```

```
 max_distance_m integer NOT NULL, -- Store the upper boundary in metres.
 upper_inclusive boolean NOT NULL DEFAULT false, -- Include only the last upper boundary.
 CHECK(min_distance_m>=0 AND max_distance_m>min_distance_m), -- Reject inverted intervals.
 EXCLUDE USING gist (numrange(min_distance_m::numeric,max_distance_m::numeric,
  CASE WHEN upper_inclusive THEN '[]' ELSE '[)' END) WITH &&) -- Reject overlapping bands.
);
CREATE TABLE nearby ( -- Map the M:N address-hawker relationship.
 flat_id bigint NOT NULL REFERENCES hdb_flat(flat_id), -- Identify the linked address.
 hawker_id integer NOT NULL REFERENCES hawker_centre(hawker_id), -- Identify the linked centre.
 distance_m numeric(9,2) NOT NULL CHECK(distance_m BETWEEN 0 AND 2000), -- Store only retained
    distances.
 PRIMARY KEY(flat_id,hawker_id) -- Prevent duplicate address-centre pairs.
);
CREATE VIEW nearby_classified AS -- Classify stored distances from the shared lookup.
SELECT n.flat_id,n.hawker_id,n.distance_m, -- Preserve each relationship key and exact distance.
 p.proximity_level_id,p.level_name AS proximity_class, -- Expose the configured band.
 p.proximity_level_id AS proximity_rank -- Keep near, medium and far in numeric order.
FROM nearby n -- Read one row per address-centre pair.
JOIN proximity_level p ON n.distance_m>=p.min_distance_m -- Include the lower boundary.
 AND (n.distance_m<p.max_distance_m -- Exclude ordinary upper boundaries.
      OR (p.upper_inclusive AND n.distance_m<=p.max_distance_m)); -- Include the final 2 km
    endpoint.
COMMIT; -- Publish the new schema without deleting any existing team schema.
```

B Data Loading SQL

```
\set ON_ERROR_STOP on
-- Run from the project root against a new database/schema: psql -f sql/00_load_psql.sql.
\i sql/01_schema.sql
SET search_path = housing, public; -- Use the newly created housing schema.
-- Bulk-load the audited town CSV using its explicit column order.
\copy housing.town (town_id,town_name) FROM 'data/processed/town.csv' WITH (FORMAT CSV,HEADER
    TRUE)
-- Bulk-load the audited hdb_flat CSV using its explicit column order.
\copy housing.hdb_flat (flat_id,town_id,block,street_name,postal_code,latitude,longitude) FROM '
    data/processed/hdb_flat.csv' WITH (FORMAT CSV,HEADER TRUE)
-- Bulk-load the audited hawker_centre CSV using its explicit column order.
\copy housing.hawker_centre (hawker_id,hawker_name,address,latitude,longitude,market_stall_count
    ) FROM 'data/processed/hawker_centre.csv' WITH (FORMAT CSV,HEADER TRUE)
-- Bulk-load the audited proximity_level CSV using its explicit column order.
\copy housing.proximity_level (proximity_level_id,level_name,min_distance_m,max_distance_m,
    upper_inclusive) FROM 'data/processed/proximity_level.csv' WITH (FORMAT CSV,HEADER TRUE)
-- Bulk-load the audited nearby CSV using its explicit column order.
\copy housing.nearby (flat_id,hawker_id,distance_m) FROM 'data/processed/nearby.csv' WITH (
    FORMAT CSV,HEADER TRUE)
-- Bulk-load the audited resale_transaction CSV using its explicit column order.
\copy housing.resale_transaction (transaction_id,flat_id,sale_month,flat_type,storey_range,
    floor_area_sqm,flat_model,lease_commence_year,remaining_lease_months,resale_price,
    source_row_key) FROM 'data/processed/resale_transaction.csv' WITH (FORMAT CSV,HEADER TRUE)
VACUUM (ANALYZE); -- Establish the same read-mostly visibility state as the benchmark.
\i sql/03_verify.sql
-- Run sql/04_queries.sql next; do not create Task 4 indexes before recording the baseline.
```

C Analytical Queries

```
SET search_path = housing, public; -- Use the merged scoped schema.
```

```sql
SELECT t.town_name, -- Identify the town.
       rt.flat_type, -- Separate 4 ROOM and 5 ROOM sales.
       round(avg(rt.resale_price/rt.floor_area_sqm),2) AS avg_price_per_sqm, -- Weight each sale
     equally.
       count(*) AS transaction_count -- Report the underlying sample size.
FROM resale_transaction rt -- Read the scoped transaction table.
JOIN hdb_flat f USING(flat_id) -- Attach the sale's address.
JOIN town t USING(town_id) -- Attach its town name.
WHERE rt.flat_type IN ('4 ROOM','5 ROOM') -- Keep the two required room types.
  AND rt.sale_month>=DATE '2023-01-01' -- Include January 2023 onward.
  AND rt.sale_month<DATE '2026-01-01' -- Include all of December 2025.
GROUP BY t.town_name,rt.flat_type -- Compute a mean for each town/type pair.
ORDER BY avg_price_per_sqm DESC,t.town_name,rt.flat_type; -- Rank groups deterministically.


WITH bucketed AS ( -- Assign each scoped sale to one remaining-lease bucket.
 SELECT f.town_id,rt.resale_price,rt.floor_area_sqm, -- Keep the values needed for aggregation.
  CASE WHEN rt.remaining_lease_months<600 THEN 1 -- Less than 50 years.
       WHEN rt.remaining_lease_months<840 THEN 2 -- At least 50 and less than 70 years.
       ELSE 3 END AS bucket_rank -- At least 70 years.
 FROM resale_transaction rt -- Read the shared three-year sample.
 JOIN hdb_flat f USING(flat_id) -- Attach the town identifier.
 WHERE rt.sale_month>=DATE '2023-01-01' -- Start the common period.
   AND rt.sale_month<DATE '2026-01-01' -- End after December 2025.
)
SELECT t.town_name, -- Name the town.
 CASE b.bucket_rank WHEN 1 THEN '< 50 years' -- Label the short-lease group.
                    WHEN 2 THEN '50 to < 70 years' -- Label the middle group.
                    ELSE '70+ years' END AS lease_bucket, -- Label the longest group.
 round(percentile_cont(0.5) WITHIN GROUP -- Use a continuous median.
  (ORDER BY b.resale_price)::numeric,2) AS median_resale_price, -- Report nominal SGD.
 round(avg(b.resale_price/b.floor_area_sqm),2) AS avg_price_per_sqm, -- Average per-sale ratios.
 count(*) AS transaction_volume -- Count each transaction once.
FROM bucketed b -- Aggregate the assigned buckets.
JOIN town t USING(town_id) -- Attach the town name.
GROUP BY t.town_name,b.bucket_rank -- Keep town and bucket populations separate.
ORDER BY t.town_name,b.bucket_rank; -- Use logical lease-bucket order.


WITH top_towns AS ( -- Rank towns using the same scoped 3 ROOM population.
 SELECT f.town_id,avg(rt.floor_area_sqm) AS avg_3room_floor_area_sqm -- Weight transaction
    observations.
 FROM resale_transaction rt -- Read in-period sales.
 JOIN hdb_flat f USING(flat_id) -- Attach the town identifier.
 WHERE rt.flat_type='3 ROOM' -- Rank only the required room type.
   AND rt.sale_month>=DATE '2023-01-01' -- Start the ranking period.
   AND rt.sale_month<DATE '2026-01-01' -- Include December 2025 in ranking.
 GROUP BY f.town_id -- Compute each town's mean area.
 ORDER BY avg_3room_floor_area_sqm DESC,f.town_id -- Break ties by a stable key.
 LIMIT 10 -- Keep exactly ten towns.
)
SELECT t.town_name, -- Identify each selected town.
 extract(year FROM rt.sale_month)::integer AS sale_year, -- Create calendar-year groups.
 round(x.avg_3room_floor_area_sqm,2) AS avg_3room_floor_area_sqm, -- Show why the town was
    selected.
 max(rt.resale_price) AS highest_resale_price -- Report the annual maximum for 3 ROOM.
FROM resale_transaction rt -- Read the scoped sale observations.
JOIN hdb_flat f USING(flat_id) -- Attach town membership.
JOIN top_towns x USING(town_id) -- Restrict to the ten selected towns.
JOIN town t USING(town_id) -- Attach town names.
WHERE rt.flat_type='3 ROOM' -- Keep the same type in the annual-maximum stage.
  AND rt.sale_month>=DATE '2023-01-01' -- Apply the same start in both stages.
  AND rt.sale_month<DATE '2026-01-01' -- Apply the same end in both stages.
GROUP BY t.town_name,sale_year,x.avg_3room_floor_area_sqm -- Keep each town/year separate.
```

```
ORDER BY x.avg_3room_floor_area_sqm DESC,t.town_name,sale_year; -- Present ranked towns and
    years.


WITH nearest AS ( -- Select one physical centre before attaching the band lookup.
 SELECT DISTINCT ON (flat_id) flat_id,hawker_id,distance_m -- Keep one nearest link per address.
 FROM nearby -- Read the original relationship instead of multiplying it by sales.
 ORDER BY flat_id,distance_m,hawker_id -- Match the useful B-tree order and break distance ties.
), prices AS ( -- Compare the same town and room type in the common date window.
 SELECT t.town_name,rt.flat_type, -- Define each comparison group.
  avg(rt.resale_price) FILTER (WHERE p.level_name='near') AS avg_near, -- Mean within 500 metres.

  avg(rt.resale_price) FILTER (WHERE p.level_name='far') AS avg_far -- Mean at 1,000-2,000
    metres.
 FROM resale_transaction rt -- Read sale observations at transaction grain.
 JOIN hdb_flat f USING (flat_id) -- Attach the sale's block address.
 JOIN town t USING (town_id) -- Attach its town name.
 JOIN nearest n USING (flat_id) -- Count each covered sale only once.
 JOIN proximity_level p ON n.distance_m>=p.min_distance_m -- Include the lower band boundary.
  AND (n.distance_m<p.max_distance_m -- Exclude ordinary upper boundaries.
       OR (p.upper_inclusive AND n.distance_m<=p.max_distance_m)) -- Include the final 2 km
    endpoint.
 WHERE rt.sale_month>=DATE '2023-01-01' -- Start the agreed three-year analysis scope.
   AND rt.sale_month<DATE '2026-01-01' -- Include all of December 2025.
 GROUP BY t.town_name,rt.flat_type -- Preserve separate comparison populations.
)
SELECT town_name AS town,flat_type, -- Match the required five-column output schema.
 round(avg_near,2) AS avg_price_near_hawker, -- Display near prices in nominal SGD.
 round(avg_far,2) AS avg_price_far_hawker, -- Display far prices in nominal SGD.
 round((avg_near-avg_far)*100/NULLIF(avg_far,0),2) AS price_diff_percentage -- Use unrounded
    means.
FROM prices -- Read the already grouped comparison values.
WHERE avg_near IS NOT NULL AND avg_far IS NOT NULL -- Require both near and far observations.
ORDER BY price_diff_percentage DESC,town,flat_type; -- Present deterministic ranked differences.


WITH eligible AS ( -- Avoid giving extra weight to multiple fresh-market centres.
 SELECT DISTINCT nc.flat_id,nc.proximity_class,nc.proximity_rank -- One address/class membership.

 FROM nearby_classified nc -- Read classified address-centre links.
 JOIN hawker_centre h USING(hawker_id) -- Attach market-stall status.
 WHERE h.has_fresh_market IS TRUE -- Exclude confirmed no and unknown status.
)
SELECT t.town_name,e.proximity_class, -- Group by town and distance band.
 round(avg(rt.resale_price),2) AS avg_resale_price, -- Average each within-group sale once.
 count(*) AS transaction_count -- Report the within-group sample size.
FROM resale_transaction rt -- Read the three-year sale observations.
JOIN hdb_flat f USING(flat_id) -- Attach town membership.
JOIN town t USING(town_id) -- Attach town names.
JOIN eligible e USING(flat_id) -- Use deduplicated fresh-market memberships.
WHERE rt.sale_month>=DATE '2023-01-01' -- Start the common analysis window.
  AND rt.sale_month<DATE '2026-01-01' -- Include the end of December 2025.
GROUP BY t.town_name,e.proximity_class,e.proximity_rank -- Keep distance groups separate.
HAVING avg(rt.resale_price)>700000 -- Use one fixed SGD 700,000 budget at every size.
ORDER BY avg_resale_price DESC,t.town_name,e.proximity_rank; -- Rank passing groups.


WITH membership AS ( -- Deduplicate small spatial memberships before joining transactions.
 SELECT DISTINCT n.flat_id,p.proximity_level_id AS proximity_rank, -- Keep one address/band key.
  p.level_name AS proximity_class, -- Use the teammate's shared band definitions.
  CASE h.has_fresh_market WHEN TRUE THEN 'yes' -- Preserve confirmed fresh-market access.
                          WHEN FALSE THEN 'no' -- Preserve confirmed absence.
                          ELSE 'unknown' END AS fresh_market_status -- Keep missing evidence
    separate.
```

```
 FROM nearby n -- Include the M:N relationship base relation.
 JOIN hawker_centre h USING (hawker_id) -- Include the centre base relation.
 JOIN proximity_level p ON n.distance_m>=p.min_distance_m -- Include the lookup base relation.
  AND (n.distance_m<p.max_distance_m -- Apply non-overlapping interval boundaries.
       OR (p.upper_inclusive AND n.distance_m<=p.max_distance_m)) -- Retain the inclusive last
    endpoint.
)
SELECT t.town_name,m.proximity_class,m.fresh_market_status, -- Define the reporting populations.
 count(DISTINCT f.flat_id) AS flat_count, -- Count participating block-address records.
 count(*) AS transaction_count, -- Count each sale once per unique membership group.
 round(avg(rt.resale_price),2) AS avg_resale_price -- Use the same population as the count.
FROM town t -- Include the town base relation.
JOIN hdb_flat f USING (town_id) -- Include the address base relation.
JOIN resale_transaction rt USING (flat_id) -- Include the sale base relation: six in total.
JOIN membership m USING (flat_id) -- Use deduplicated memberships, avoiding hawker fan-out.
WHERE rt.sale_month>=DATE '2023-01-01' -- Start the same period used by Q1-Q5.
  AND rt.sale_month<DATE '2026-01-01' -- Include all three complete calendar years.
GROUP BY t.town_name,m.proximity_class,m.proximity_rank,m.fresh_market_status -- Separate
    statuses.
ORDER BY t.town_name,m.proximity_rank,m.fresh_market_status; -- Preserve logical distance order.
```

D Index Statements

```
SET search_path = housing, public; -- Apply the measured four-index bundle.

CREATE INDEX idx_rt_type_month
ON resale_transaction (flat_type, sale_month)
INCLUDE (flat_id, floor_area_sqm, resale_price); -- Cover Q1 and both Q3 scans.

CREATE INDEX idx_rt_flat
ON resale_transaction (flat_id)
INCLUDE (transaction_id, resale_price); -- Support flat-led transaction lookups in Q4-Q6.

CREATE INDEX idx_nearby_flat_dist
ON nearby (flat_id, distance_m, hawker_id); -- Supply Q4's nearest-centre order.

CREATE INDEX idx_nearby_hawker
ON nearby (hawker_id)
INCLUDE (flat_id, distance_m); -- Cooperate with idx_rt_flat to remove Q6's spilling sort.

VACUUM (ANALYZE) resale_transaction; -- Equalise visibility and refresh planner statistics.
VACUUM (ANALYZE) nearby; -- Refresh statistics for both proximity access paths.

-- Deliberately omitted: an index on remaining_lease_months. Q2 classifies every
-- transaction with CASE and therefore cannot use that index to eliminate rows.
```

E Benchmark SQL

```
SET search_path = housing, public;

DROP INDEX IF EXISTS idx_transaction_type_month;
DROP INDEX IF EXISTS idx_transaction_remaining_lease;
DROP INDEX IF EXISTS idx_proximity_block_distance;
DROP INDEX IF EXISTS idx_hawker_fresh_market;

DROP TABLE IF EXISTS benchmark_results;
CREATE TABLE benchmark_results (
```

```
  size_level integer NOT NULL,
  transaction_rows bigint NOT NULL,
  query_no integer NOT NULL,
  repetition integer NOT NULL,
  indexed boolean NOT NULL,
  execution_ms numeric(14,3) NOT NULL
);

CREATE TEMP TABLE benchmark_transaction_stage (
 block varchar(16), street_name varchar(120), sale_month date, flat_type varchar(20),
 storey_range varchar(20), floor_area_sqm numeric(7,2), flat_model varchar(60),
 lease_commence_year smallint, remaining_lease_months integer,
 resale_price numeric(12,2), source_row_key char(64));

CREATE TEMP TABLE benchmark_queries (query_no integer PRIMARY KEY, sql_text text NOT NULL);
INSERT INTO benchmark_queries VALUES
(1, $q$SELECT t.town_name, rt.flat_type, avg(rt.resale_price/rt.floor_area_sqm), count(*)
 FROM resale_transaction rt JOIN hdb_flat f USING(flat_id) JOIN town t USING(town_id)
 WHERE sale_month >= (SELECT max(sale_month)-interval '3 years' FROM resale_transaction)
 GROUP BY t.town_name,rt.flat_type$q$),
(2, $q$SELECT t.town_name, width_bucket(rt.remaining_lease_months,0,1188,3),
 percentile_cont(0.5) WITHIN GROUP(ORDER BY rt.resale_price), avg(rt.resale_price/rt.
    floor_area_sqm),count(*)
 FROM resale_transaction rt JOIN hdb_flat f USING(flat_id) JOIN town t USING(town_id)
 GROUP BY t.town_name,width_bucket(rt.remaining_lease_months,0,1188,3)$q$),
(3, $q$WITH top_towns AS (SELECT f.town_id FROM resale_transaction rt JOIN hdb_flat f USING(
    flat_id)
 WHERE rt.flat_type='3 ROOM' GROUP BY f.town_id ORDER BY avg(rt.floor_area_sqm) DESC LIMIT 10)
 SELECT t.town_name,extract(year from rt.sale_month),max(rt.resale_price)
 FROM resale_transaction rt JOIN hdb_flat f USING(flat_id) JOIN town t USING(town_id)
 WHERE f.town_id IN (SELECT town_id FROM top_towns)
 GROUP BY t.town_name,extract(year from rt.sale_month)$q$),
(4, $q$WITH nearest AS (SELECT DISTINCT ON(flat_id) flat_id,proximity_level_id
 FROM v_flat_hawker_proximity ORDER BY flat_id,distance_m)
 SELECT t.town_name,rt.flat_type,avg(rt.resale_price) FILTER(WHERE n.proximity_level_id=1),
 avg(rt.resale_price) FILTER(WHERE n.proximity_level_id=3)
 FROM resale_transaction rt JOIN hdb_flat f USING(flat_id) JOIN town t USING(town_id)
 JOIN nearest n USING(flat_id) GROUP BY t.town_name,rt.flat_type$q$),
(5, $q$SELECT t.town_name,p.level_name,avg(rt.resale_price),count(DISTINCT rt.transaction_id)
 FROM resale_transaction rt JOIN hdb_flat f USING(flat_id) JOIN town t USING(town_id)
 JOIN v_flat_hawker_proximity p USING(flat_id)
 JOIN hawker_centre h USING(hawker_id) WHERE h.has_fresh_market IS TRUE
 GROUP BY t.town_name,p.level_name HAVING avg(rt.resale_price)>200000$q$),
(6, $q$SELECT t.town_name,p.level_name,h.has_fresh_market,count(DISTINCT f.flat_id),
 count(DISTINCT rt.transaction_id),avg(rt.resale_price)
 FROM town t JOIN hdb_flat f USING(town_id) JOIN resale_transaction rt USING(flat_id)
 JOIN flat_hawker_proximity fp USING(flat_id) JOIN proximity_level p
 ON fp.distance_m>=p.min_distance_m AND (fp.distance_m<p.max_distance_m
 OR (p.upper_inclusive AND fp.distance_m<=p.max_distance_m))
 JOIN hawker_centre h USING(hawker_id)
 GROUP BY t.town_name,p.level_name,h.has_fresh_market$q$);

CREATE OR REPLACE PROCEDURE run_six_queries(p_size integer, p_indexed boolean)
LANGUAGE plpgsql AS $$
DECLARE r record; i integer; started timestamptz; ignored bigint; n bigint;
BEGIN
  SELECT count(*) INTO n FROM resale_transaction;
  FOR r IN SELECT * FROM benchmark_queries ORDER BY query_no LOOP
    EXECUTE 'SELECT count(*) FROM (' || r.sql_text || ') result' INTO ignored; -- warm-up
    FOR i IN 1..5 LOOP
      started := clock_timestamp();
      EXECUTE 'SELECT count(*) FROM (' || r.sql_text || ') result' INTO ignored;
      INSERT INTO benchmark_results VALUES
        (p_size,n,r.query_no,i,p_indexed,extract(epoch FROM clock_timestamp()-started)*1000);
```

```
    END LOOP;
  END LOOP;
END $$;

CREATE OR REPLACE PROCEDURE load_benchmark_size(p_size integer)
LANGUAGE plpgsql AS $$
DECLARE file_path text;
BEGIN
  file_path := U&'C:/Users/BUDINGLIN/OneDrive/Desktop/\5B66\4E1A\8BFE\7A0B/NTU MSC DS 2026 -
    2027/2026 Semester 1/SD6103 Data Systems/SD6103 Data Systems Group Project/data/processed/
    resale_transaction_size'
              || p_size || '.csv';
  TRUNCATE resale_transaction, benchmark_transaction_stage RESTART IDENTITY;
  EXECUTE format('COPY benchmark_transaction_stage FROM %L WITH (FORMAT csv, HEADER true,
    ENCODING ''UTF8'')',file_path);
  INSERT INTO resale_transaction
   (flat_id,sale_month,flat_type,storey_range,floor_area_sqm,flat_model,lease_commence_year,
    remaining_lease_months,resale_price,source_row_key)
  SELECT f.flat_id,s.sale_month,s.flat_type,s.storey_range,s.floor_area_sqm,s.flat_model,s.
    lease_commence_year,
         s.remaining_lease_months,s.resale_price,s.source_row_key
  FROM benchmark_transaction_stage s JOIN hdb_flat f USING(block,street_name);
  ANALYZE resale_transaction;
  CALL run_six_queries(p_size,false);
END $$;

CALL load_benchmark_size(1);
CALL load_benchmark_size(2);
CALL load_benchmark_size(3);
CALL load_benchmark_size(4);
CALL load_benchmark_size(5);

CREATE INDEX idx_transaction_type_month ON resale_transaction(flat_type,sale_month)
 INCLUDE(flat_id,floor_area_sqm,resale_price);
CREATE INDEX idx_transaction_remaining_lease ON resale_transaction(remaining_lease_months)
 INCLUDE(flat_id,resale_price,floor_area_sqm);
CREATE INDEX idx_proximity_block_distance ON flat_hawker_proximity(flat_id,distance_m)
 INCLUDE(hawker_id);
CREATE INDEX idx_hawker_fresh_market ON hawker_centre(hawker_id) WHERE has_fresh_market IS TRUE;
ANALYZE;
CALL run_six_queries(5,true);

COPY (
 SELECT size_level,transaction_rows,query_no,indexed,
        round(avg(execution_ms),3) AS mean_ms,
        round(percentile_cont(0.5) WITHIN GROUP(ORDER BY execution_ms)::numeric,3) AS median_ms,
        round(stddev_samp(execution_ms),3) AS sd_ms
 FROM benchmark_results GROUP BY size_level,transaction_rows,query_no,indexed
 ORDER BY indexed,size_level,query_no
) TO 'C:/Users/BUDINGLIN/Desktop/SD6103 Data Systems Project/report/generated/benchmark_summary.
    csv'
WITH (FORMAT csv,HEADER true);

SELECT indexed,count(*) AS observations,min(execution_ms),round(avg(execution_ms),3),max(
    execution_ms)
FROM benchmark_results GROUP BY indexed ORDER BY indexed;
```

F Reproducibility Commands

```
powershell -File scripts/download_data.ps1
```

```
python scripts/prepare_data.py
psql -U postgres -f sql/00_create_database.sql
psql -U postgres -d sd6103_housing -f sql/01_schema.sql
psql -U postgres -d sd6103_housing -f sql/02_load_data.sql
psql -U postgres -d sd6103_housing -f sql/03_queries.sql
```